# Geopredictor Weighting Review: MC's Scripts vs. the Live Pipeline

**Date:** 2026-07-26 **Scope:** Steps 3-5 of the snvdem-col pipeline (CDF geopredictor data + expert relevance weights -> combining weights with geopredictors), plus the diagnostic mapping that checks both. **Files reviewed:**

- MC's scripts (`drafts/1025-v3/Coppedge/geopredictors/`): `weighting_July2026corrected.R`, `wts_predictors_v2.R`, `ColMaps.R`, and `MemoCalculations.docx` (MC's own written reasoning for the "July" formula in `wts_predictors_v2.R`, discussed in Section B).
- Live pipeline counterparts: `04_vdem_data/02_vdem_weighting/02_vdem_weighting.R`, `05_weighting/01_weighting` `06_benchmark/02_diagnostics/01_scripts/01_validate_map.R`
- Also updated: `03_geocoded_panel/01_clean_geocoded/01_clean_geocode.R` (Step 3), source of the geopredictor direction-flip relocated into Step 5.

**Section A** (expert relevance weights): the live pipeline matches MC's own formula. **Section B** (combining geopredictors with weights): the live pipeline has been **updated to implement MC's "July" formula** (`MemoCalculations.docx`/`wts_predictors_v2.R`'s `Indices_JulyMC`), replacing the prior formula (what his script calls "MPSA version"). Everything below reflects the pipeline as it stands after this update. **Section C:** Summary of changes to pipeline in recent weeks.

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## A. V-Dem Expert Relevance Weights (Step 4)

**MC's script: `weighting_July2026corrected.R` (1,693 lines)**

**Produces** `ELCLweights_wide.dta` — one row per country-year, with 44 columns (`el_Rural…el_None_of_the_above`, `cl_Rural…cl_None_of_the_above`). Each column is a "relevance weight": how much more often V-Dem coders cited that criterion (e.g. "Rural", "Indigenous", "Remote") to explain why a subnational area was MORE vs. LESS free-and-fair (or stronger/weaker on civil liberties). This is the file for describing how much each geographic predictor should count for a given country-year.

**Structure:**

- **Lines 1-16:** loads V-Dem's coder-level dataset from MC's personal Dropbox (`C:/Users/mcoppedg/Dropbox/...`), computes `year` from `historical_date`, and de-duplicates to Dec-31-only rows (coders' year-end stamping; Jan-1 rows are duplicates of the prior year's coding).
- **Lines 18-61.** This block is uncommented prose describing indicator names, concept notes, and a design sketch for the weighting formula. (No leading `#`, so the script can't be run top-to-bottom as a single batch job.)
- **Lines 62-165:** builds the *elections* HPD-range calculation (`snlsffHPD`) — per country-year, how far a municipality's score could plausibly sit above/below the national V-Dem estimate, weighted by how many coders reported subnational variation. Includes a recent fix (2026-07-03) to the category-weight direction.
- **Lines 168-247:** the same range calculation for *civil liberties* (`SNHPD`/`wtdCL_range`), built from a proxy mean (`CLSNmean = v2x_civlib * (100 - v2clsnlpct)/100`) since V-Dem has no direct subnational CL summary variable.
- **Lines 249-1560:** the relevance-weight calculation itself, repeated by hand once per criterion (0-21) per variable family (`v2elsnmrfc`/`v2elsnlfc` for elections, `v2clrgstch`/`v2clrgwkch` for civil liberties) — roughly 90 nearly-identical copy-pasted blocks (e.g. `snlfc_0` through `snlfc_21`, then the same shape again for `snmrfc_*`, `snwkcc_*`, and a fourth family for "stronger CL" not shown above but structurally identical). Each block counts how many coders picked each 0/1 answer for that criterion, per country-year. These get cbind'd together (by column position, not by key) and turned into proportions (`pr_0…pr_21`), then the "more/stronger" and "less/weaker" proportion tables are merged and differenced (`abs(more - less)`) to get the final relevance weight per criterion.
- **Line 1560-1563:** merges the elections and CL weight tables into `ELCLweights_wide` and writes it out.

**Fixes present in this version:** the criterion-weight zero-count bug (criteria 2/3/4 dividing by criterion 1's zero-count instead of their own) and the pr_5/pr_6 (Outside_capital/North) column-order swap are both corrected here, applied consistently across all 4 variable families.

**Caveats:** personal Dropbox paths throughout (not portable), and per above, not runnable as a single batch script.

**Live pipeline: `04_vdem_data/02_vdem_weighting/02_vdem_weighting.R` (417 lines)**

Same target output (`ELCLweights_wide.dta`, in the `03_outputs/` folder) and same underlying formulas as MC's script, restructured as reusable functions instead of ~90 hand-copied blocks:

- **`count_wide()` (lines 81-89):** the one function that replaces every `snlfc_i`/`snmrfc_i`/ `snwkcc_i`/`snstcc_i`-style block in MC's script — pivots one coder-level 0/1 column to country-year wide counts, using a proper `count()`/`pivot_wider()` instead of manual grouping.
- **`compute_criterion_weights()` (lines 275-312):** replaces the whole "22 blocks x 4 prefixes" section — takes one of the 4 raw variable-family prefixes and returns all 22 criterion proportions per country-year in one call. Includes the `FIX_KNOWN_BUGS` switch (line 51, default `TRUE`) that reproduces MC's two confirmed bugs on demand (denominator using criterion 1's zero-count for criteria 2/3/4; the pr_5/pr_6 swap) for audit purposes.
- **`more_minus_less()` (lines 331-339):** joins the more/less (or stronger/weaker) tables **by criterion name**, not column position.
- **Lines 91-247:** the elections/CL HPD-range sections — same fixed formula as MC's script (category-weight direction corrected 2026-07-03), plus a plotting fix: `coord_cartesian()` instead of `scale_y_continuous(limits=...)` so out-of-range points get clipped visually rather than silently dropped as missing rows.
- **Lines 341-357:** builds `el_wide`/`cl_wide`/`ELCLweights_wide` with both fixes applied.
- **Lines 409-416:** writes `ELCLweights_wide.dta` to `03_outputs/` (the pre-fix version is preserved separately as `ELCLweights_wide_ORIGINAL_pre-fix_2026-07-02.dta` for audit).

**Comparison: Expert Relevance Weights**

| Aspect | MC's script | Live pipeline |
|---|---|---|
| Formula correctness | Both known bugs fixed | Same fixes, via `FIX_KNOWN_BUGS` switch |
| Result | — | **Identical** for COL: 125 country-years x 44 columns, max abs diff = 0.0 |
| Code structure | ~90 hand-copied blocks | 2 reusable functions |
| Merge method | Positional `cbind()` (fragile) | Key-based `full_join`/`merge` |
| Portability | Personal Dropbox paths, not batch-runnable | Shared-drive paths, runs via `Rscript` |

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## B. Combining Geopredictors with Weights (Step 5)

**MC's script: `wts_predictors_v2.R` (539 lines)**

**Produces:** per-municipality-year `snelect` (electoral) and `sncivlib` (civil liberties) index scores for Colombia, by taking the 16 geocoded predictor columns (urbanization, distance to capital, remoteness, etc.) and computing a weighted average using the relevance weights from Section A.

**Structure:**

- **Lines 1-16:** loads `ELCLweights_wide.dta` and `CDF_averages_v1.rds` (both read from local copies MC downloaded, not the shared drive), builds `wt_el_1011`/`wt_cl_1011` by averaging the Civil_unrest and Illicit_activity weight columns into one.
- **Lines 29-49:** checks the median of each predictor (mostly ~0.5, except South/East, which are single directional predictors rather than symmetric pairs) and builds hi/lo split columns for the five paired predictors (urban/rural, development, distance-to-capital, unrest/illicit, ruling-party) at the 0.5 threshold.
- **Lines 51-165:** exploratory analysis — correlation scatterplots between predictor pairs, a Spearman-correlation check on an "extreme rural" subset against a second, differently-named input file (`imputed_cdf_panel07_26.rds`), and quadrant-identification of extreme municipalities. None of this feeds the final index; it's scratch work.
- **Line 170-171:** the actual join used downstream — `Indices <- predict_hilo %>% left_join(weights_col, by = "year")`.
- **Lines 173-333:** more diagnostics — correlation matrices and 12 scatterplots checking that each predictor's spatial pattern looks sensible.
- **Lines 335-372 — "Indices_JulyMC":** computes `snelect`/`sncivlib` by flipping *geopredictor values*, not weights, for Ruling_party and North – `(1-avg15t16hi)` for Ruling_party_strong (`avg15t16lo` unflipped for Ruling_party_weak; see `MemoCalculations.docx`), and `(1-avg6)` for North – while leaving Indigenous, Ruling_party, and Inside/Outside_capital *weights* unflipped (raw).
- **Lines 374-409** — "Indices_both"/"MPSA version" (the pipeline's prior formula, now superseded): computed `snelect`/`sncivlib` by instead flipping the *weight* for exactly 5 criteria (Indigenous, Ruling_party_strong, Ruling_party_weak, Inside_capital, Outside_capital) via `1 - el_Indigenous` etc.
- **Line 412:** `write_rds(Indices_JulyMC, "Indices_both.rds")` — the "JulyMC" result is saved under the filename "Indices_both.rds," even though a differently-computed object literally named `Indices_both` (the MPSA version) exists later in the same script — a naming trap for anyone loading `Indices_both.rds` expecting the MPSA formula.
- **Lines 414-525:** more diagnostics and choropleth/bubble maps.
- **Lines 529-533:** the MPSA-version `Indices_both` gets `sndem = 0.5*(snelect+sncivlib)` added and is written to `08_final_snvdem_data/MC/SNDEM_tentative.rds` — MC's own script's final output uses the MPSA version, not the JulyMC version saved to `Indices_both.rds` at line 412.

`MemoCalculations.docx`

MC offered the "July"/"Indices_JulyMC" formula in his latest memo:

- **States plainly that the prior approach was wrong:** *"This code corrects errors in the previous version that a) subtracted weights rather than geopredictors from 1 and b) was based on a different understanding of how some of the geopredictor variables were aligned with democracy."*
- **Urban, More_development, Inside_capital** ("the most straightforward geopredictors"): the variable is already aligned with democracy in the right direction once the single Step 3/Step 5 geopredictor flip is applied, so no further weight or geopredictor adjustment is needed.
- **Civil_unrest/Illicit_activity, Sparse_population, Remote, Indigenous:** labeled as if negatively related to democracy, but *the underlying variables* are actually already positively related once flipped – e.g. "Indigenous population in effect actually measures White or Mestizo population." So these also need no further adjustment beyond the one geopredictor flip.
- **Ruling_party (strong/weak):** explicitly designed as a *non-linear, inverted-V* relationship – below 50% ruling-party vote share, more support helps democracy (genuine popularity); above 50%, more support hurts it (overwhelming dominance). This is *why* only the "strong" (>50%) side gets its geopredictor flipped and the "weak" side doesn't.
- **North/South/East/West:** the memo describes a two-part correction, but flags its own premise as unverified — the memo's derivation ends with "`[Check this with CDF_averages7_20_26.rds]`." Using the updated data, the reversal MC described is not present, which leaves only his theoretical claim about North (independent negative relationship with democracy) unconfirmed either way. Live pipeline leaves North, South, East, and West all unflipped, same as before this review, pending MC's

direct confirmation on North specifically – the earlier same-day version of this pipeline flipped North on a re-derivation from his claim, but that was reverted per PM rather than act on a claim MC hasn't separately confirmed post-data-fix.
- **Cumaribo note:** the memo separately flags a likely miscoding of Cumaribo ("the largest municipality in the country") based on its facet maps. PM addressed this: it was a DIVIPOLA geocoding error that has since been corrected.

**Updated live pipeline**

`05_weighting/01_weighting_geopredictors/01_weighting_geopredictors.R` was updated to implement the above. The prior version is preserved as `01_weighting_geopredictors_ORIGINAL_pre-julyformula_2026-07-26.` and its output as `03_output/snvdem_col_weighted_ORIGINAL_pre-julyformula_2026-07-26.rds`.

- **Lines 111-143 (geopredictor direction-flip block):** North, South, East, and West are all left unflipped, same as before this update. A same-day version briefly added `avg6 = 1 - avg6` (North), re-derived from MC's theoretical claim now that the data-reversal premise behind his memo's literal derivation no longer holds – reverted per PM: implementing a re-derivation rather than something MC stated directly isn't the right call without his confirmation. North remains an open question, exactly like South/East/West.
- **Lines 159-168 (hi/lo split):** added `avg15t16hi_flip = 1 - avg15t16hi`, implementing the Ruling_party inverted-V. This column is always defined (equal to 1 when a row is outside the "strong" regime), reproducing MC's `(1-avg15t16hi)` term exactly – it is the other half of one continuous tent-shaped function together with `avg15t16lo`.
- **Line 232 (`pred_cols`):** Ruling_party_strong's slot now points at `avg15t16hi_flip` instead of raw `avg15t16hi`.
- **Lines 253-261 (`el_wt_mat/cl_wt_mat`):** removed the `1 - weight` flip for Indigenous, Ruling_party_strong, Ruling_party_weak, Inside_capital, and Outside_capital – these now use the raw relevance weight, matching MC's July formula. Direction for these criteria is now carried entirely by the geopredictor side (already handled above), not the weight.
- **`weighted_avg_narm()` and its 50%-weight-mass floor are unchanged** – this renormalization logic (MC's formula doesn't have an equivalent) is layered on top of the new formula. NAs remain at 0 (previously 682 with a naive sum, 48 before this renormalization was added on 2026-07-03).

**Connection to the paper's theoretical claims**

Two of the criteria this update touches already have a stated position in the paper, and MC's memo lines up with both:

- **Ruling_party dominance:** the paper argues the *machine-politics* interpretation — a dominant ruling party signals capture/patronage, not popularity, and should count against the democracy score. MC's memo's inverted-V (dominance above 50% hurts democracy) is consistent with this.
- **Capital proximity:** the paper argues that proximity to Bogota signals more state capacity, and should count toward the democracy score. MC's memo groups "inside capital" among the "most straightforward" geopredictors, positively associated with democracy with no further adjustment needed — also consistent.
- **Indigenous population:** no stated paper position, and the memo's own justification (avg14 "in effect actually measures White or Mestizo population," assumed positive) is a substantive claim that should be agreed to by the team (secondary literature available?).

**By the numbers: what changed when the live pipeline adopted MC's July formula**

Compared the live pipeline's actual output before and after the formula update, using the same live inputs (`ELCLweights_wide.dta`, `CDF_averages.rds`) in both cases, so every difference below comes from the formula change alone. n = 26,928 municipality-years (COL, 2000-2023).

**Overall distribution:**

| Statistic | snelect — prior (MPSA) | snelect — current (July, North unflipped) | sncivlib — prior | sncivlib — current | sndem — prior | sndem — current |
|---|---|---|---|---|---|---|
| Min | 0.111 | 0.077 | 0.126 | 0.112 | 0.121 | 0.099 |
| Median | 0.333 | 0.385 | 0.354 | 0.380 | 0.344 | 0.382 |
| Mean | 0.331 | 0.379 | 0.353 | 0.373 | 0.342 | 0.376 |
| Max | 0.535 | 0.615 | 0.584 | 0.590 | 0.556 | 0.580 |

The current (July) formula runs uniformly higher — every quartile shifted up. `weighted_avg_narm()`'s renormalization means neither version has any NAs (0 in both), even though MC's own formula alone (without that renormalization) would reintroduce 634.

**Row-level difference (current minus prior), same municipality-year matched 1:1:**

| Index | Mean diff | Mean abs diff | Max abs diff | Correlation |
|---|---|---|---|---|
| `snelect` | +0.048 | 0.066 | 0.216 | 0.747 |
| `sncivlib` | +0.021 | 0.042 | 0.144 | 0.823 |
| `sndem` | +0.034 | 0.052 | 0.147 | 0.797 |

A correlation of 0.71-0.82 means this is a visibly different measure of subnational democracy. Municipality-years reorder, not just shift.

**Rank stability** (Spearman correlation of `sndem` within each year — i.e. if only relative ranking of municipalities mattered, not absolute level):

| Year | Spearman rho |
|---|---|
| 2000 | 0.777 |
| 2008 | 0.889 |
| 2016 | 0.895 |
| 2023 | 0.879 |

Even in the best year (2016), roughly 1 - 0.895^2, about 20%, of rank variance differs between the two formulas — non-trivial for anything that compares municipalities to each other rather than tracking one municipality over time.

**National year-means for `sndem`** (the number that actually appears in a published trend line):

| Year | Prior (MPSA) mean | Current (July, North unflipped) mean | Diff |
|---|---|---|---|
| 2000 | 0.359 | 0.356 | -0.004 |
| 2005 | 0.326 | 0.346 | +0.020 |
| 2008 | 0.313 | 0.374 | +0.060 |
| 2010 | 0.314 | 0.390 | +0.076 |
| 2013 | 0.319 | 0.397 | +0.078 |
| 2016 | 0.356 | 0.386 | +0.029 |
| 2020 | 0.357 | 0.389 | +0.032 |
| 2023 | 0.371 | 0.380 | +0.009 |

The gap is not constant over time: near-zero in 2000, widening to roughly +0.08 in 2010-2013, then narrowing again toward 2023. The formula change also changes the shape of the time trend.

**Where the change concentrates** (top single-row differences, `snelect`):

| MPIO_CDPMP | Municipality | Year | Prior (MPSA) | Current (July) | Diff |
|---|---|---|---|---|---|
| 25489 | Nimaima (Cundinamarca) | 2010 | 0.323 | 0.539 | +0.216 |
| 25489 | Nimaima (Cundinamarca) | 2009 | 0.315 | 0.530 | +0.216 |
| 25489 | Nimaima (Cundinamarca) | 2011 | 0.316 | 0.529 | +0.213 |
| 25489 | Nimaima (Cundinamarca) | 2008 | 0.308 | 0.520 | +0.212 |
| 25489 | Nimaima (Cundinamarca) | 2012 | 0.318 | 0.530 | +0.212 |

The largest single-row changes (~0.21, on a [0,1]-ish scale) land on the same handful of municipalities repeated across years – Nimaima especially – consistent with those municipalities having unusually extreme values on exactly the criteria the formula change touches (Indigenous, Ruling_party, Capital), rather than a diffuse shift spread evenly everywhere. North is no longer part of what changed – see below.

**North/South/East/West: reconsidered, reverted to unflipped.** A same-day version of this update briefly flipped North (re-derived from MC's theoretical claim, now that the data reversal behind his memo's literal derivation is verified gone). That was reverted per PM: implementing a re-derivation of MC's claim isn't the same as implementing something he stated directly, and shouldn't go live without his confirmation. North, South, East, and West are all currently unflipped – identical to the pipeline's treatment before today. The isolated effect of the North question alone (measured before the revert, flipping North only vs. leaving it unflipped, holding everything else fixed) was correlation 0.98, mean abs diff ~0.014 – a real but secondary effect next to the Indigenous/Ruling_party/Capital change, and one now absent from the live pipeline until MC confirms.

**Facet map: pre-benchmarked `sndem`, 2000-2023**

The central-highlands "hotspot" pattern MC's memo describes ("better democracy in the central highlands, worse outcomes on the borders and coastlines, especially in the south") is visible here. Cumaribo still sticks out but this is probably reflective of its position in the index.

**Comparison: Combining Geopredictors with Weights**

| Aspect | MC's script | Live pipeline |
|---|---|---|
| Which formula | "MPSA version" (superseded) vs. "Indices_JulyMC" (current) — MC's own script's tail end still uses MPSA, per `MemoCalculations.docx` this is inertia, not intent | **Now implements "Indices_JulyMC"**, per `MemoCalculations.docx` |
| 5-of-16 weight-flip (Indigenous, Ruling_party x2, Capital x2) | Present in "MPSA" only | **Removed 2026-07-26** — per MC's memo, this was the error his July formula corrects |
| Ruling_party inverted-V | Present in "Indices_JulyMC" via geopredictor flip | **Implemented 2026-07-26** via `avg15t16hi_flip` |

| Aspect | MC's script | Live pipeline |
|---|---|---|
| North vs. South/East/West flip | Shipped code flips North; memo's literal derivation flips South/East/West | **Unflipped** (all four) – briefly implemented a North-only flip same-day, reverted per PM pending MC's direct confirmation |
| Missing-predictor handling | None — one missing criterion blanks the whole row | `weighted_avg_narm()`: renormalizes over available criteria, floors at 50% weight mass |

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## C. Key Changes to the Live Pipeline (last few weeks)

- **2026-07-02:** Criterion-weight zero-count bug found and fixed (criteria 2/3/4 dividing by criterion 1's zero-count instead of their own).
- **2026-07-03:**
  - HPD range category-weight inversion fixed (elections and CL both previously weighted "no variation" coders by 2x instead of "significant variation" coders — understated the range in most country-years).
  - `01_weighting_geopredictors.R`'s plain weighted-sum found to blank whole rows on any single missing predictor (a side effect of Step 2's `na.last="keep"` fix letting real NAs flow through); decided and implemented `weighted_avg_narm()` with a 50%-weight-mass floor — NAs 682 -> 48 (see `narm_decision_log_2026-07-03.md`).
  - `04_vdem_data` and `05_weighting` reorganized into numbered subfolders (source-files/script+MC-legacy/outputs/images), mirroring `01_empirical_data`'s structure.
  - Positional `cbind()` merges replaced with key-based joins in the criterion-weight calculation (protects against silent misalignment if criteria gain uneven country-year coverage).
- **2026-07-06:**
  - `scale_y_continuous(limits=...)` (silently drops out-of-range rows before plotting) replaced with `coord_cartesian(ylim=...)` (zooms only, never drops data) in both HPD-range ribbon plots, after a visible gap in the Colombia 1948-1953 range confirmed the bug.
  - Confirmed directly (not just by inference) that MC's legacy range formula produces impossible negative ranges for 40% of all country-year observations globally — documented so old figures built from that formula aren't mistaken for a correctness reference.
  - Output-file locations cleaned up: canonical outputs (`ELCLweights_wide.dta`, `snvdem_col_weighted.rds`) promoted out of legacy `MC/` folders into each step's own `03_outputs/03_output`; unbenchmarked and benchmarked data no longer share one folder.
  - Removed a since-obsolete manual NA-fill hardcode for `cl_Less_development` (2000-2002/2004) once the underlying merge/weight fixes made it a no-op.
  - Persisted the civil-unrest/illicit-activity scatter plot via `ggsave()` (previously only displayed in interactive sessions, silently dropped in batch runs).
- **2026-07-21:** Documented and finalized the missing-predictor renormalization design (`weighted_avg_narm()` + `MIN_WEIGHT_FRACTION`) as the pipeline's committed approach.
- **2026-07-26:** Relocated the geopredictor direction-flip (avg0t1, avg4t5, avg10t11, avg14, avg15t16, and the avg13 blend) from Step 3 (`01_clean_geocode.R`'s "Ensure correct score directions" section) to Step 5 (`01_weighting_geopredictors.R`, immediately after loading `CDF_averages.rds`), so direction is decided in exactly one place instead of two. `CDF_averages.rds` now stores raw (pre-flip) values for those columns; `avg13`'s two raw components (`lAR_13pkm`/`DisMer_13`) are no longer merged in Step 3 either, since flipping after averaging would incorrectly flip both components instead of just the one that needs it. Caught and fixed one regression along the way: `ViolInd_1011`'s raw name lacks the "t" its siblings have, so the generic rename-to-`avg<N>` step would have silently produced `avg1011` instead of `avg10t11` — added an explicit rename. Validated end-to-end (0.0 max abs diff, 26,928/26,928 rows

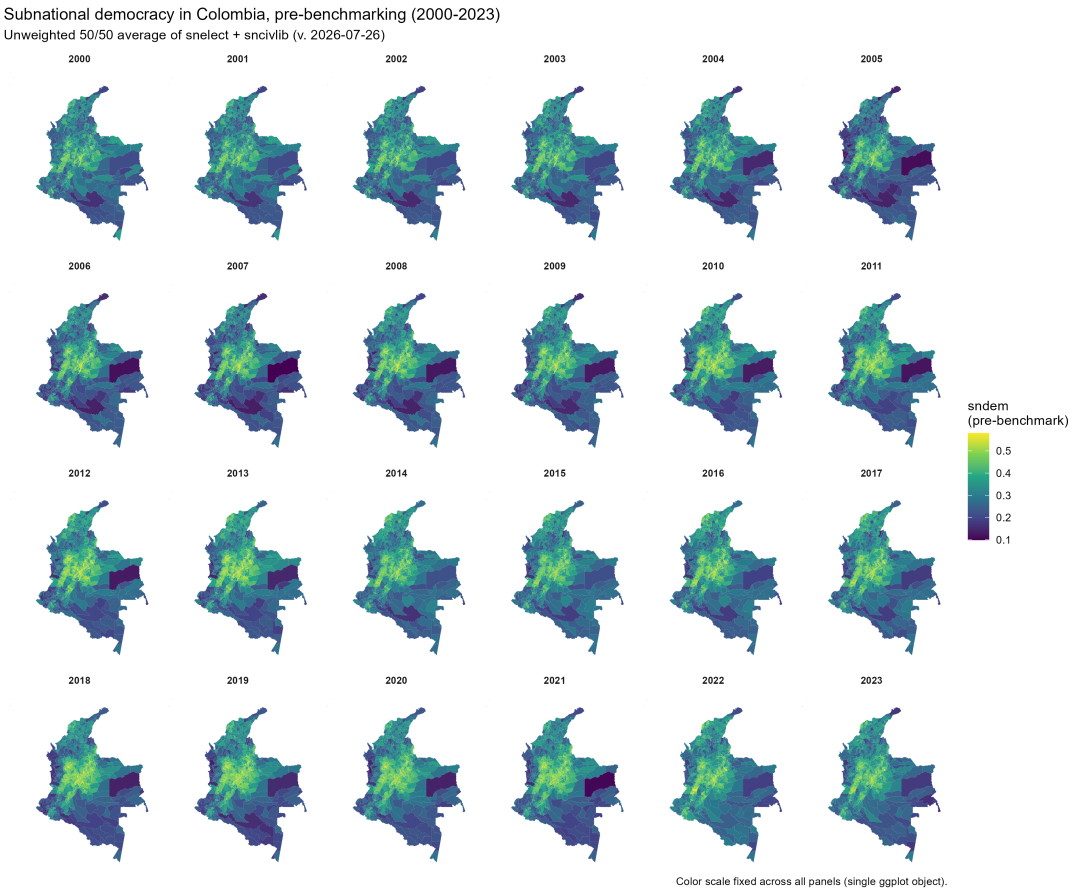



Figure 1: Subnational democracy (pre-benchmark), 2000-2023 – July formula

matched, identical NA pattern) before applying to the live paths.

- **2026-07-26:** Found and read MC's `MemoCalculations.docx`, his written reasoning for the "July" formula in `wts_predictors_v2.R`. Verified his cardinal-direction reasoning (North/South/East/West) against current data: the raw-data reversal his memo describes is not present in the current `CDF_averages.rds` (spot-checked Apartado, Cucuta, Cali, Puerto Leguizamo against known geography; national `cor(avg6,avg7) = -0.71`, `cor(avg8,avg9) = -0.65`, both consistent with correct, non-reversed coding).
- **2026-07-26:** Updated `01_weighting_geopredictors.R` to implement MC's July formula: added a geopredictor flip for North (re-derived from his theoretical claim against verified current data); added the Ruling_party inverted-V via a new `avg15t16hi_flip` column; removed the `1 - weight` flip for Indigenous, Ruling_party_strong/weak, and Inside/Outside_capital in `el_wt_mat/cl_wt_mat`. Validated against independently-computed versions of MC's formula before applying to the live paths. Prior script version preserved as `01_weighting_geopredictors_ORIGINAL_pre-julyformula_2026-07-26.R`; prior output preserved as `snvdem_col_weighted_ORIGINAL_pre-julyformula_2026-07-26.rds`. `weighted_avg_narm()` and its missing-data renormalization are unchanged by this update.
- **2026-07-26:** Reverted the North flip added earlier the same day. Per PM: implementing a re-derivation of MC's theoretical claim about North (rather than something he stated directly, now that the data-reversal premise behind his literal memo derivation is verified gone) shouldn't go live without his confirmation. North, South, East, and West are all unflipped again, matching the pipeline's treatment before this review started. Intermediate (North-flipped) output preserved as `snvdem_col_weighted_ORIGINAL_pre-northrevert_2026-07-26.rds`; live output and the pre-benchmark facet map regenerated to match.

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*Prepared for internal review and for sharing with MC. Questions on any specific line range above should reference the file and line numbers cited, which correspond to the versions of each script as reviewed on 2026-07-26.*